CONCEPT MAP

3SFP6I

Gaseous State

IDEAL GASES

obey following gas laws under all conditions

1. Boyle's Law $P \alpha \frac{1}{V}$ (T, n constant)
2. Charles Law - $V \alpha T$ (P, n constant)
3. Gay Lusaac - $P \alpha T$ (V, n constant)
4. Avogadro's Hypothesis - $V \alpha n$ (T, P constant)
5. Ideal gas equation $PV = nRT$

REAL GASES

obey Van der Waals equation

$$\left(P + \frac{a n^2}{V^2}\right)(V - nb) = nRT$$

↓

Critical constant

$$T_C = \frac{8a}{27 R b}, \quad P_C = \frac{a}{27 b^2}$$

and $V_C = 3b$

Gases

- **Diffusion** → Graham's Law of diffusion
- **Critical phenomenon** → Critical temperature, Critical volume, Critical pressure
- **Liquefaction of gases** → Joule-Thomson effect

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